On-Orbit SWIR Imagery MTF Compensation for the Geology-1 Satellite Using Cross-Spectral Structural Priors

Haixu He, Jining Yan, *Senior Member, IEEE*, Zichen Jing, and Lizhe Wang, *Fellow, IEEE*

***Abstract*—The Geology-1 hyperspectral satellite employs a lightweight and highly integrated payload architecture, establishing a novel paradigm for commercial small-satellite-based geological observation. However, the compact optomechanical configuration exacerbates the coupling among residual attitude disturbances, cryocooler-induced microvibrations, and the imaging chain, leading to systematic degradation of the system modulation transfer function (MTF) in short-wave infrared (SWIR) imagery. To address this challenge, we propose a structure-guided, two-stage Geology-1 SWIR imagery MTF compensation (GSMC) framework. This framework leverages coregistered visible and near-infrared (VNIR) imagery, which offers superior spatial resolution and enhanced imaging stability, as a natural structural prior. Consequently, a zero-shot compensation model is constructed without requiring external training data. Experimental results demonstrate that GSMC delivers consistent and superior performance across diverse geological scenes. Quantitative evaluation shows that the MTF at the Nyquist frequency increases from 0.204 to 0.297 after compensation. The proposed method preserves high radiometric fidelity, with a mean relative error of only 0.57%, while also improving downstream task performance by increasing land-cover classification accuracy from 83.43% to 89.19%. In addition, spectral consistency analysis verifies that the cross-spectral guidance strategy does not distort SWIR-specific information, with spectral angle mapper (SAM) values remaining below 0.05. These results indicate that the proposed framework effectively compensates for time-varying jitter degradation in Geology-1 imagery and provides reliable technical support for quantitative on-orbit processing and high-precision geological mapping.**

***Index Terms*—Blind compensation, Geology-1, modulation transfer function (MTF) compensation, short-wave infrared (SWIR).**

I. INTRODUCTION

IN RECENT years, driven by the widespread adoption of commercial off-the-shelf (COTS) space components and reduced launch barriers [1], [2], small satellites (<500 kg) have overcome the traditional constraints of high development costs and lengthy development cycles in the aerospace sector [3]. Leveraging their abbreviated development cycles, reduced costs, and unique capability to achieve high-frequency Earth revisit through constellation deployment, small satellites have rapidly emerged as a transformative trend in space industry development [4]. For example, the SuperDove satellites (5.8 kg) deployed by Planet Labs Inc., enable daily global land-surface coverage through constellation networking [5]. Similarly, the Jilin-1 High Resolution-03 series satellites (43 kg), developed by Chang Guang Satellite Technology Company Ltd., provide high-resolution optical images with a spatial resolution of 0.75 m, demonstrating the remarkable balance achieved by small satellites between lightweight design and high performance [6].

As a representative hyperspectral small satellite, Geology-1 has a total mass of 88.3 kg and overcomes the limitations of hyperspectral payloads, which are typically bulky and costly [7]. The satellite employs a lightweight off-axis three-mirror anastigmat optical system with a compact design integrating visible and near-infrared (VNIR) and short-wave infrared (SWIR) detectors. Through spectral splitting, deep multifocal-plane integration is achieved, enabling the VNIR and SWIR detection modules to be accommodated within an extremely constrained payload volume (Fig. 1). Concurrently, the highly

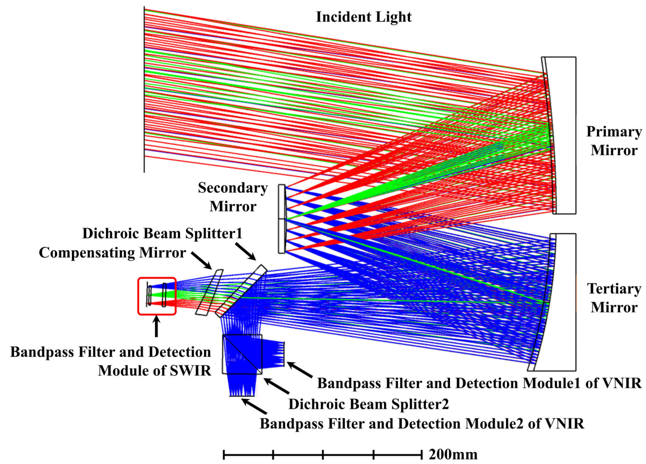

Fig. 1. Optical system configuration of the Geology-1 satellite.

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integrated and miniaturized architecture significantly reduces intercomponent spacing, resulting in intensified coupling among attitude control residual disturbances, microvibrations from mechanical components, and the imaging chain. Even minor vibrations can induce jitter in optical elements, thereby degrading imagery quality [8], [9], [10], [11]. Furthermore, the SWIR detector relies on a mechanical cryocooler to suppress dark current noise, whose high-frequency microdisturbances are readily transmitted to the focal plane within the limited structural volume. When these disturbances are superimposed with small pitch fluctuations from attitude control, perceptible along-track image smear is likely to occur [12], resulting in degradation of the system modulation transfer function (MTF) [13].

In satellite imagery MTF compensation, the processing pipeline is typically divided into two tightly coupled stages: point spread function (PSF) estimation and MTF compensation [14]. From a system perspective, MTF compensation is not merely a perceptual enhancement problem, but a physically grounded process of compensating for MTF degradation through explicit or implicit PSF inversion. Collectively, these stages determine both restoration quality and practical engineering feasibility [15]. In the PSF estimation stage, due to the randomness, nonstationarity, and rapid temporal variation of on-orbit microvibrations [16], the PSF is often unknown and time-varying. Consequently, traditional kernel estimation methods based on static priors are difficult to apply directly [17]. Research on satellite imaging generally exploits two categories of information sources. The first comprises blind kernel estimation based solely on imaging data, where PSFs are inferred using natural imagery gradient sparsity, edge consistency, or images of ground calibration targets [18], [19], [20]. The second incorporates physical priors derived from onboard auxiliary measurements, such as attitude jitter data from inertial measurement units, gyroscopes, and star trackers. By mapping platform microvibration models into convolution kernels, this approach enhances the physical interpretability of kernel estimation [9], [21]. Once the PSF is obtained, the process proceeds to the MTF compensation stage, whose objective is to maximally recover ground-feature details based on the estimated PSF [22]. In frequency-domain terms, this process can be equivalently interpreted as compensating for the attenuation of the system MTF induced by the PSF. Unlike natural imagery restoration, optical satellite imagery is constrained by noise levels, illumination dynamic range, and spectral bandwidth, placing greater emphasis on robustness and controllability [23]. Classical nonblind deconvolution methods, such as Wiener filtering and the Lucy–Richardson method, are widely adopted in onboard processing chains due to their stable physical interpretations and engineering reliability [24], [25]. In recent years, nonblind methods integrating deep learning have also emerged, introducing prior regularization while preserving the physical meaning of the PSF to suppress ringing artifacts and over-sharpening [26], [27].

Beyond the serial "PSF-first, compensate-later" framework, joint optimization strategies have also been developed, in which the latent sharp image and the time-varying PSF are simultaneously estimated within a unified optimization framework [28], [29], [30], [31], [32]. By imposing tighter imaging constraints, these methods minimize the propagation of PSF estimation errors into the final restoration results. They exploit the intrinsic image-prior properties of network architectures to adaptively disentangle microvibration blur from image content during optimization, thereby enabling joint recovery of both the image and the PSF [29]. For time-varying and difficult-to-model blur caused by on-orbit microvibrations, joint estimation offers a more principled way to handle PSF uncertainty and generally yields improved robustness in weak-texture regions or under high-noise conditions. Recent advances in dynamic and probabilistic kernel modeling have further strengthened the capability of unsupervised blind restoration [30], [31], [32]. In particular, dynamic kernel-prior models adaptively learn kernel priors through Markov chain Monte Carlo (MCMC) sampling over stochastic kernel distributions [30]. Kernel distribution learning methods characterize kernel uncertainty probabilistically within alternating optimization frameworks tailored to hyperspectral and multispectral data [31]. Moreover, MCMC-based kernel approximation combined with metalearning incorporates network-level Langevin dynamics to mitigate entrapment in poor local optima [32]. Although these dynamic and probabilistic approaches are theoretically appealing for handling spatially or temporally varying blur, the proposed Geology-1 SWIR imagery MTF compensation (GSMC) framework adopts a deterministic formulation tailored to the practical requirements of satellite on-orbit processing, where computational efficiency and physical interpretability are of primary importance.

Despite significant theoretical progress in MTF compensation research, multiple practical challenges remain when applying these methods to the data processing pipeline of the Geology-1 satellite. First, due to the 30-m spatial resolution of the SWIR imagery, MTF measurement methods based on standard ground targets are difficult to implement effectively. Moreover, onboard microvibrations exhibit randomness and nonstationarity, resulting in time-varying PSFs. In addition, it is nearly impossible to acquire strictly spatiotemporally paired "degraded–sharp" benchmark datasets, rendering it difficult to directly apply deep learning methods that rely heavily on large amounts of supervised data.

To address these challenges, we propose a GSMC framework that exploits the platform characteristic of coaligned VNIR–SWIR imaging. Owing to its higher spatial resolution, superior imaging stability, and the absence of cryocooler-induced microvibrations, the VNIR imagery serves as a natural structural prior for SWIR MTF compensation. We design a decoupled two-stage optimization pipeline comprising PSF estimation and MTF compensation. First, guided by the VNIR imagery, the blind deconvolution process is constrained to regions with distinct ground textures, enabling robust PSF estimation. Subsequently, under a fixed-PSF assumption, constrained nonblind MTF compensation is performed to jointly ensure spatial detail recovery and radiometric consistency. Without relying on any external training data, the proposed framework achieves high-quality SWIR imagery restoration while simultaneously preserving radiometric accuracy and geological structural fidelity.

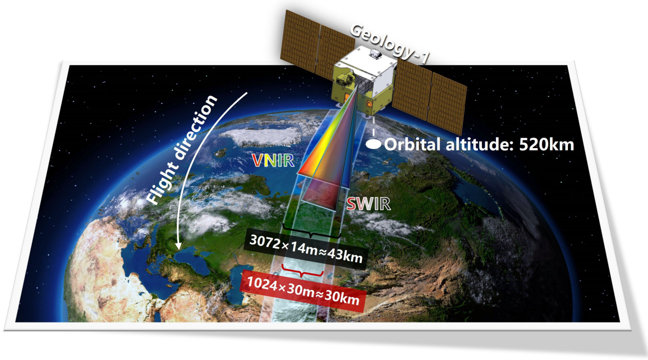

Fig. 2. Schematic illustration of the VNIR and SWIR sensor imaging geometry on the Geology-1 satellite.

II. Material and Method

A. Geology-1 Satellite

The Geology-1 hyperspectral satellite is equipped with both VNIR and SWIR detectors. Under identical orbital conditions, the two detectors perform coordinated imaging, producing images with a common along-track centerline. However, they exhibit differences in spatial resolution and swath width, as illustrated in Fig. 2. In particular, the VNIR detector provides a spatial resolution of 14 m and a swath width of approximately 43 km, offering superior spatial detail and broader coverage. The SWIR detector has a spatial resolution of 30 m, an imaging swath of approximately 30 km, and comprises ten spectral bands covering the wavelength range from 1020 to 2480 nanometers (nm). The proposed method is applied to the SWIR data after image tiling. For each tiled single-scene SWIR image, the number of pixels in the cross-track direction is 1024. In the along-track direction, the image extent is matched to the VNIR imaging swath to construct an approximately square spatial coverage, resulting in approximately 1400 pixels, corresponding to an along-track length of about 43 km.

Given the inherent differences between the VNIR and SWIR detectors in terms of swath width and spatial resolution, rigorous spatial harmonization preprocessing is first performed on both types of imagery. In particular, geometric coregistration between VNIR and SWIR images is conducted using the open-source AROSICS registration library [33], and the VNIR images are resampled to match the spatial resolution of the SWIR data. Subsequently, the VNIR images are adaptively cropped using the effective imaging area of the SWIR data as a reference. Through these steps, a spatially aligned and scale-consistent clear VNIR guidance image z_g and its corresponding SWIR observation image z_b are constructed, providing standardized inputs for GSMC.

B. Problem Definition and Framework

Image degradation in satellite imaging processes can typically be modeled as a convolution operation. From the perspective of physical imaging mechanisms, such degradation mainly originates from the spatial response of the imaging system to an ideal point source—the PSF—which comprehensively characterizes the combined effects of the optical system, detector integration, and satellite vibrations on imaging quality. In algorithmic modeling, the PSF is commonly represented in an equivalent discrete form as a convolution kernel. Let the spectral band image $I_B \in \mathbb{R}^{H\times W}$ denote the observed degraded image. The degradation model is defined as

$$I_B = I_S \otimes k + n$$

where I_S represents the latent sharp image, k denotes an unknown jitter-induced PSF that approximates the spatial diffusion effects caused by time-varying vibrations during the actual imaging process, $\otimes$ denotes the convolution operator, and n represents additive white Gaussian noise. The objective of this study is not generic image restoration for visual enhancement, but rather MTF compensation under realistic on-orbit degradation conditions. In this formulation, the estimated PSF k serves as a discrete spatial-domain representation of the system PSF, whose Fourier transform directly corresponds to the degraded MTF. Therefore, restoring I_S under a physically constrained PSF k is equivalent to compensating for the frequency-domain attenuation imposed by the imaging system.

Since both I_S and k are unknown, blind image restoration is inherently a severely ill-posed inverse problem [34]. Direct optimization is highly susceptible to convergence to local minima, often yielding solutions with significant uncertainty and instability [35]. From a statistical inference perspective, this problem can be formulated as a maximum a posteriori (MAP) estimation

$$\min_{I_S,k} \frac{1}{2}\|I_S \otimes k - I_B\|_2^2 + \Phi(I_S) + \Psi(k)$$

where $\|\cdot\|_2^2$ denotes the data fidelity term, and $\Phi(I_S)$ and $\Psi(k)$ represent regularization terms imposed on the image and the PSF, respectively, to alleviate the ill-posedness of the problem and stabilize the optimization process.

In this study, we employ the parameter space of deep neural networks to implicitly represent $\Phi(I_S)$ and $\Psi(k)$, effectively constraining the solution space through the generative capacity of the networks. In particular, a dual-generator network is first utilized to construct a joint optimization mechanism for the PSF and an intermediate latent image, enabling robust estimation of an equivalent PSF. Subsequently, with the optimal PSF fixed, targeted nonblind MTF compensation is performed (Fig. 3). The core of this framework lies in the deep integration of endogenous radiometric priors and external cross-sensor structural guidance across different optimization stages. This integration effectively narrows the feasible solution space, thereby ensuring the stability of PSF estimation and achieving high-fidelity image restoration under complex and time-varying degradation conditions.

C. Stage 1: Joint PSF Estimation via Dual Generative Networks

The objective of the first stage is to robustly estimate an equivalent PSF that characterizes SWIR MTF degradation. Coregistered VNIR imagery is employed as a structural prior

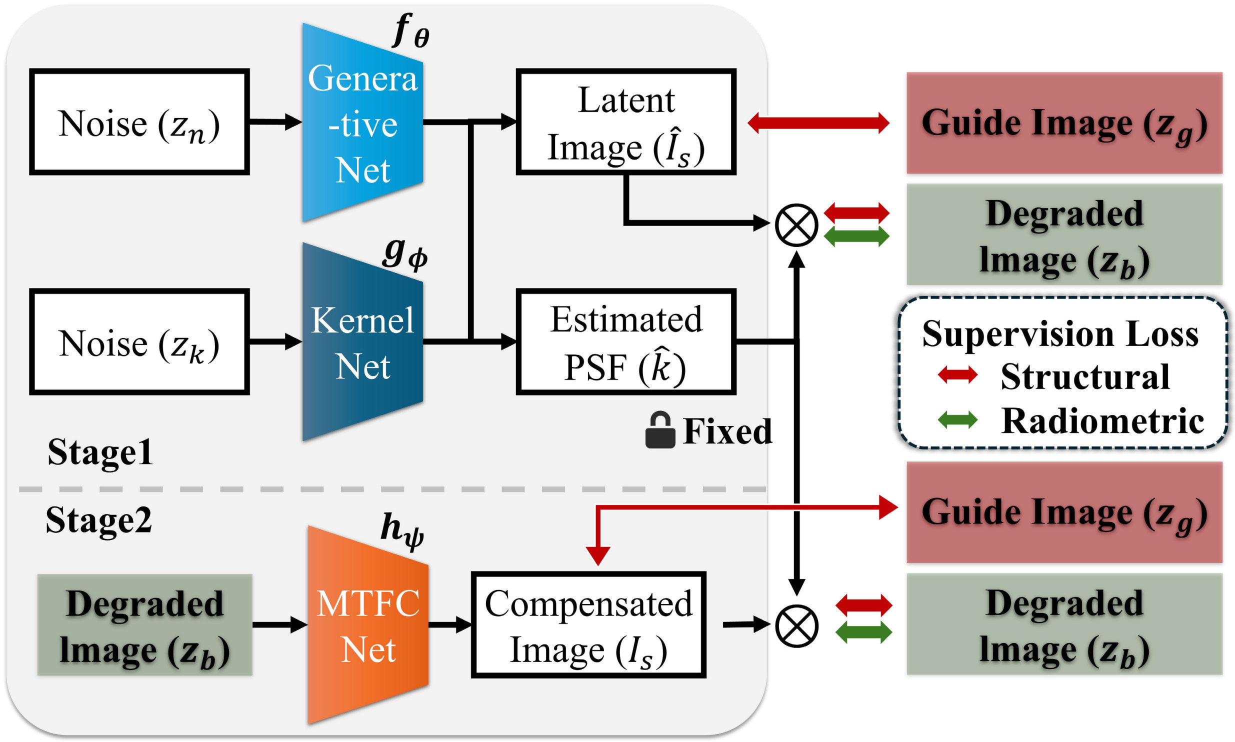

Fig. 3. Training pipeline of the proposed two-stage GSMC framework.

due to its superior spatial fidelity and enhanced imaging stability, enabling robust PSF estimation without external training data. Inspired by SelfDeblur [28], we construct two collaboratively optimized generative networks: an image generation network f_θ and a PSF generation network g_ϕ, defined as

$$\hat{I}_S = f_\theta(z_n), \quad \hat{k} = g_\phi(z_k) \tag{1}$$

where z_n denotes a uniformly distributed noise tensor input to f_θ for generating the latent image $\hat{I}_S$ and z_k is a 1-D noise vector input to g_ϕ to produce the corresponding PSF estimate $\hat{k}$. Since the noise inputs themselves contain no structural information from the observed imagery, the networks are unable to directly replicate degradation patterns. Consequently, during parameter optimization, image and PSF priors are implicitly learned through the network architecture and optimization dynamics, enabling effective decoupling between the two.

From a physical imaging perspective, the PSF must satisfy energy conservation and nonnegativity constraints [36]. To this end, a Softmax mapping operator $\mathcal{S}(\cdot)$ is introduced at the output of the PSF generation network g_ϕ to normalize the network output into a valid probability distribution

$$\hat{k}_i = \mathcal{S}\left(g_\phi(z_k)_i\right) = \frac{\exp\left(g_\phi(z_k)_i\right)}{\sum_j \exp\left(g_\phi(z_k)_j\right)} \tag{2}$$

thereby explicitly enforcing

$$\sum_i \hat{k}_i = 1, \quad \hat{k}_i \geq 0. \tag{3}$$

To ensure that the generated latent image and PSF consistently explain the degradation observed in the imagery, a reconvolution consistency constraint is adopted. In particular, the convolution of the generated sharp image $\hat{I}_S$ with the estimated PSF $\hat{k}$ is enforced to approximate the observed SWIR image z_b

$$z_b \approx \hat{I}_S \otimes \hat{k}. \tag{4}$$

However, relying solely on this reconstruction constraint is prone to yielding trivial identity solutions. To further constrain the solution space, the coregistered VNIR image z_g is introduced as a cross-spectral structural prior. The structural similarity (SSIM) is adopted to measure cross-spectral structural consistency. By decomposing image similarity into luminance, contrast, and structural components, SSIM enables the model to emphasize structural correspondence while reducing sensitivity to absolute intensity differences. This property allows the proposed GSMC framework to exploit structural information from the VNIR imagery while suppressing the influence of cross-spectral intensity discrepancies on the SWIR compensation process. By integrating the above constraints, the joint optimization objective for Stage 1 is formulated as

$$\begin{aligned}\mathcal{L}_{Stage1} = &\underbrace{\left\|\hat{I}_S \otimes \hat{k} - z_b\right\|_2^2}_{\text{Radiation Consistency}} \\ &\underbrace{+ \lambda_1 \mathcal{L}_{SSIM}\left(\hat{I}_S \otimes \hat{k}, z_b\right) + \lambda_2 \mathcal{L}_{SSIM}\left(\hat{I}_S, z_g\right)}_{\text{Texture Consistency}}\end{aligned} \tag{5}$$

where $\mathcal{L}_{SSIM}(\cdot)$ denotes the structural similarity loss function, and λ_1 and λ_2 are weighting coefficients that balance the relative importance of each constraint.

Guided by reconvolution-based radiometric consistency and VNIR-derived cross-spectral structural priors, the image and PSF generation networks achieve effective decoupling during optimization, yielding a physically consistent and

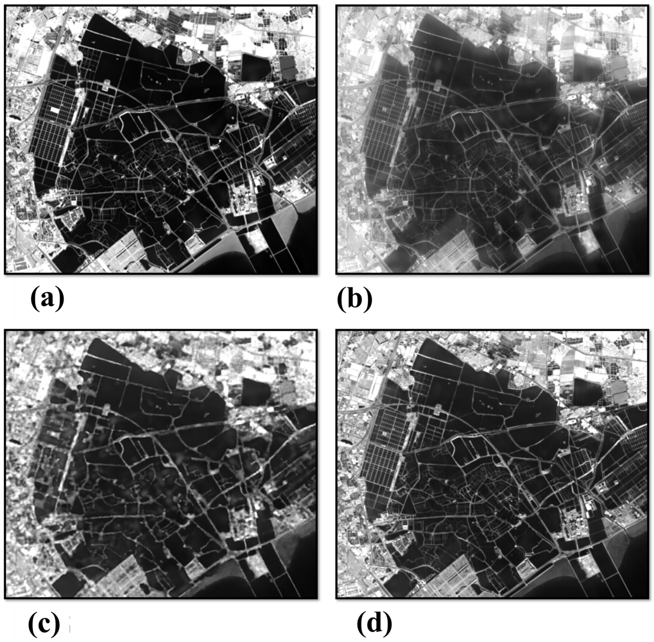

Fig. 4. Illustration of intermediate and final outputs. (a) VNIR guide image, (b) degraded SWIR image, (c) intermediate result $\hat{I}_s$ from Stage 1, and (d) final compensated image I_s from Stage 2.

stable PSF estimate $\hat{k}$ that supports subsequent nonblind MTF compensation.

D. Stage 2: High-Fidelity MTF Compensation With Fixed PSF

Limited by the representational characteristics of deep image priors, the images produced at this stage remain deficient in fine textures and high-frequency details and may be accompanied by artifacts. For remote sensing imagery, critical information such as lithological boundaries, linear structures, and fine-grained geomorphological textures often relies heavily on high-frequency components [37]. Once such information is degraded by accumulated blur, it becomes difficult to accurately recover through a noise-driven reconstruction process, as illustrated in Fig. 4(c).

Therefore, in this stage, we introduce a network h_ψ specifically designed for the nonblind MTF compensation task to further recover stable and refined high-frequency structures. The optimal PSF $\hat{k}$ estimated in the first stage is fixed as a known prior, thereby transforming the problem from an ill-posed blind deconvolution into a well-constrained nonblind deconvolution problem, which can be expressed as

$$I_S = h_\psi \left(z_b \mid \hat{k}, z_g \right). \tag{6}$$

The MTF compensation network h_ψ directly takes the observed image z_b as input. This design enables the network to fully exploit the low-frequency radiometric information and global spatial structure preserved in the original imagery, while focusing on compensating for the high-frequency information loss induced by jitter under the constraint of a fixed PSF. Since the PSF is no longer treated as an optimization variable, the solution space of the deconvolution problem is significantly reduced, effectively preventing structural drift and unstable convergence.

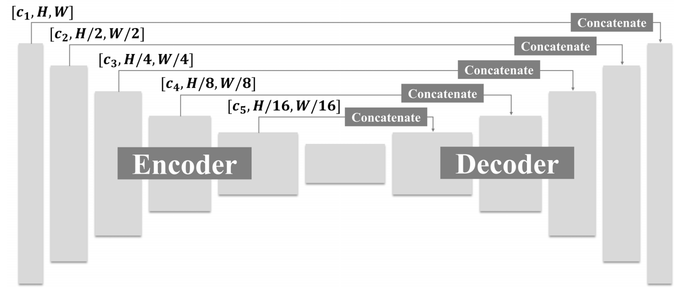

Fig. 5. U-Net architecture employed in the image generation and MTF compensation networks.

During optimization, reconvolution consistency is retained as the core constraint, requiring that the compensated image, when convolved with the fixed PSF $\hat{k}$, can consistently explain the observed image. Meanwhile, a structural consistency constraint derived from the VNIR image is introduced to further enhance detail recovery. The overall loss function for Stage 2 is defined as

$$\begin{aligned}\mathcal{L}_{Stage2} = {} & \underbrace{\left\| I_S \otimes \hat{k} - z_b \right\|_2^2}_{\text{Radiation Consistency}} \\ & + \underbrace{\lambda_1 \mathcal{L}_{SSIM}\left(I_S \otimes \hat{k}, z_b \right) + \lambda_2 \mathcal{L}_{SSIM}\left(I_S, z_g \right)}_{\text{Texture Consistency}}. \end{aligned} \tag{7}$$

The second stage effectively compensates for the high-frequency information that is difficult to recover in the first stage [Fig. 4(d)]. This significantly enhances the structural integrity and detail fidelity of the remote sensing imagery, yielding high-quality restoration results suitable for subsequent fine-scale interpretation and quantitative analysis.

E. Network Architecture

For the PSF estimation network g_ϕ, considering that its output is a 1-D PSF vector used to represent the along-track PSF, the network does not need to model complex spatial contextual relationships. Therefore, a multilayer perceptron (MLP) architecture composed of multiple fully connected layers is sufficient. Both the image generation network f_θ and the MTF compensation network h_ψ adopt a U-Net architecture, as illustrated in Fig. 5. This architecture employs an encoder–decoder design combined with skip connections, enabling simultaneous modeling of global semantic information and local detail features across multiple scales [38].

Although f_θ and h_ψ serve different optimization objectives at different stages, with the former generating latent sharp images to assist PSF estimation and the latter performing high-fidelity nonblind MTF compensation under a fixed PSF, their network architectures are kept identical. This design choice helps avoid inconsistencies in feature distributions caused by structural differences, thereby ensuring continuity in representational capacity across the two stages and enhancing both the stability of the overall optimization process and the interpretability of results.

F. Implementation Details

1) Model Configuration: In the experiments, the maximum blur displacement is set to 50 pixels to encompass the possible vibration amplitudes of the satellite under different operational conditions. Regarding the network architecture configuration, as shown in Fig. 5, the number of feature channels at each scale level of the U-Net, denoted as C_1, C_2, C_3, C_4, C_5, is set to $16, 32, 64, 128, 256$, respectively. For the PSF estimation network, the hidden-layer dimension of the multilayer perceptron is uniformly set to 1000. In the loss function design, both λ_1 and λ_2 are set to 1.

2) Adaptive Iteration Control Strategy: To enhance the stability of the optimization process and avoid unnecessary over-iteration, a windowed early stopping mechanism is designed. In Stage 1 (PSF estimation), the mean change rate of the PSF $\hat{k}$ is continuously monitored over a sliding window of 50 consecutive iterations. When the improvement rate falls below 0.1%, the PSF is considered to have converged to a physically steady state, and the system automatically terminates the iteration and fixes the current optimal PSF parameters. In Stage 2 (nonblind MTF compensation), the convergence behavior of the reconstruction loss function $\mathcal{L}_{Stage2}$ is monitored. If the mean change rate of the loss function within a sliding window of the same length (50 iterations) falls below 0.5%, the image restoration process is deemed to have stabilized, and the optimization is stopped.

3) Training Environment: All algorithmic modules are implemented in Python based on the PyTorch deep learning framework, with the Adam optimizer uniformly employed during model optimization. The experimental computing platform is equipped with 40 GB of memory and a GeForce RTX 4080 GPU. Under this hardware configuration, the joint optimization process for a single scene typically converges stably within 1500 iterations, achieving a favorable balance between image restoration accuracy and overall computational efficiency.

III. Result

A. Comparison With Other Algorithms

A total of 80 scenes was selected to evaluate the proposed method. The B6 band in the SWIR was used as the experimental spectral band, while the B15 band was selected as the VNIR guidance band. For quantitative evaluation, five representative scenes were further chosen from the dataset. To validate the effectiveness of the proposed approach, it was compared with several representative methods, including Wiener deconvolution [39], SelfDeblur [28], DIPDKP [30], W-DIP [29], and IMTF [40]. Among them, Wiener is a nonblind deconvolution method, for which GSMC provides the estimated PSF. SelfDeblur, DIPDKP, and W-DIP are deep-learning-based blind image restoration methods that jointly estimate the latent sharp image and the PSF, and their reconstruction networks adopt the same hidden-layer dimensionality as that used in this work. IMTF estimates the PSF using the knife-edge technique and performs MTF compensation via inverse convolution with a hyper-Laplacian prior.

1) Visual Comparison: Fig. 6 presents the MTF compensation results of different methods on degraded imagery. Scene Fig. 6(a) corresponds to a desert area in Xinjiang; scenes (b)–(d) were acquired over Tianjin and represent mountainous, cropland, and port areas, respectively; scene (e) depicts a mining area in Gansu. In the scenes, as shown in Fig. 6(a) and (b), terrain textures in the original degraded images are degraded, with ridge lines and gullies appearing indistinct. When processing these natural landform scenes, GSMC and Wiener exhibit relatively strong texture recovery capabilities. On the contrary, SelfDeblur, DIPDKP, and W-DIP tend to produce excessive smoothing in regions with complex textures, resulting in the loss of high-frequency details, such as fine folds in mountainous areas. Although IMTF recovers some high-frequency information, it introduces pronounced ringing artifacts, which hinder the reliable interpretation of geological details. In the cropland scene in Fig. 6(c) and the port scene in Fig. 6(d), the ground objects contain abundant linear edges and geometric structures. In the degraded images, field boundaries, road grids, and port wharf edges are noticeably diffused. The GSMC method not only effectively suppresses blur but also accurately restores furrow details in croplands and the edge contours of port infrastructures. In particular, in scene Fig. 6(d), the land–water boundaries and the edges of wharf facilities appear the sharpest in the GSMC results, with ghosting and blur largely eliminated. By contrast, Wiener deconvolution further aggravates blur and ghosting artifacts, which also demonstrates the effectiveness of the Stage 2 MTF compensation network in our framework. In the mining scene, as shown in Fig. 6(e), which contains numerous high-reflectance bright targets, GSMC preserves natural brightness transitions while enhancing fine details through effective prior constraints, resulting in the most visually favorable performance in subjective evaluation.

2) Quantitative Comparison: Relying solely on visual inspection is often insufficient to accurately quantify performance differences among various imagery restoration algorithms. Therefore, we employ widely recognized image quality assessment metrics for quantitative comparison. The signal-to-noise ratio (SNR) measures the relative proportion between effective signal components and noise, reflecting an algorithm's capability in noise suppression. The root-mean-square (rms) value is closely related to image contrast and grayscale dynamic range. The edge strength level (ESL) characterizes the sharpness of image edges. The corresponding mathematical definitions can be found in [40].

Table I presents the quantitative evaluation results across five scenes. Overall, GSMC demonstrates clear advantages in key metrics reflecting image sharpness and structural fidelity. First, in terms of the ESL metric, GSMC achieves the highest values in all scenarios, indicating that GSMC can significantly enhance edge responses of ground objects, making high-frequency structures such as ridgelines and field ridges more distinct. Second, with respect to the rms metric, GSMC either attains the leading performance or yields values slightly below the best-performing method. Since MTF degradation typically compresses the grayscale distribution and reduces image contrast, a higher rms value suggests that the proposed

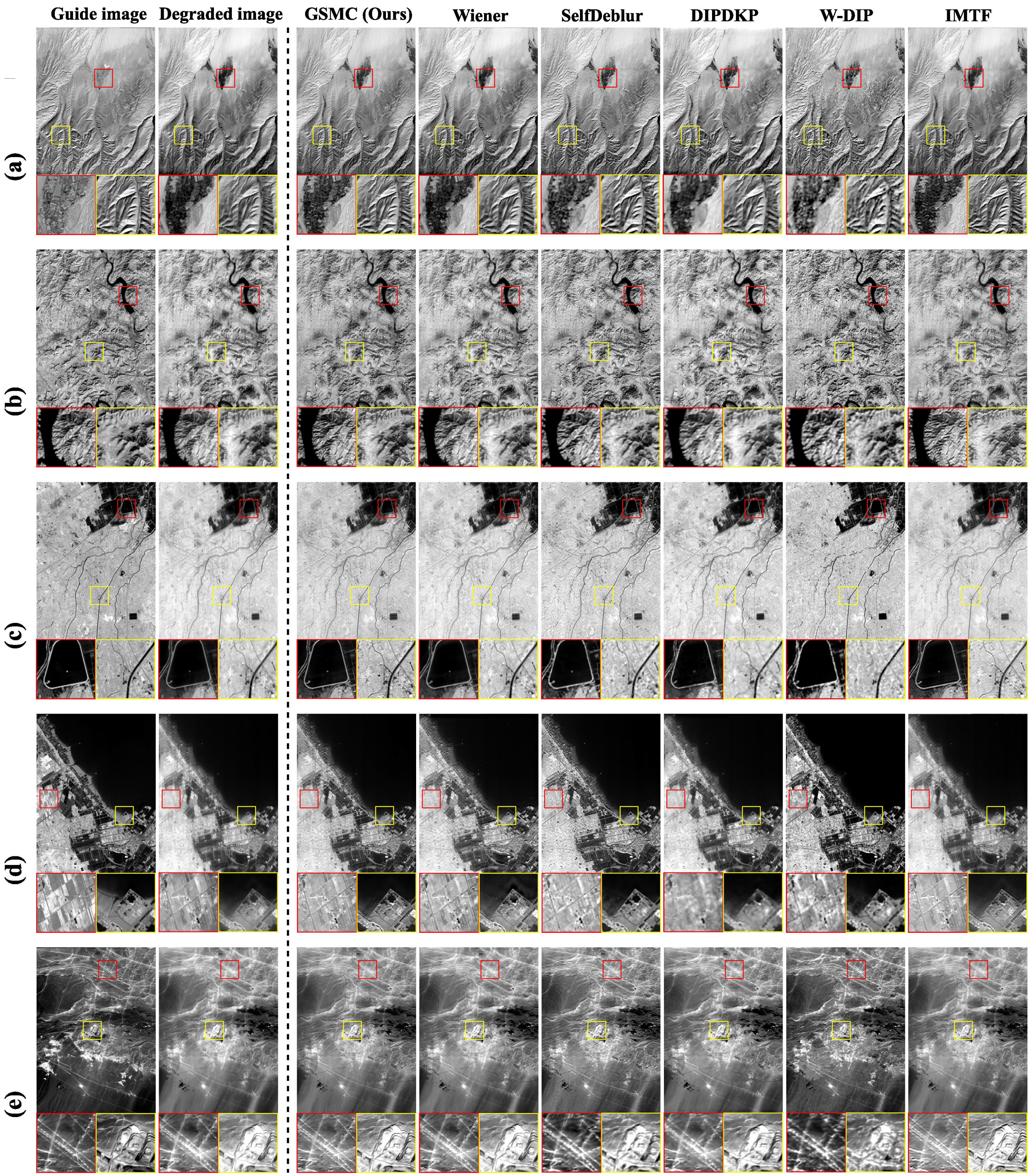

Fig. 6. Visual comparison of MTF compensation results across five representative scenes. (a) Scene 1. (b) Scene 2. (c) Scene 3. (d) Scene 4. (e) Scene 5.

method effectively restores the grayscale dynamic range. It is worth noting that, for the SNR metric, the DIPDKP and W-DIP methods achieve relatively higher scores. This phenomenon primarily stems from the strong implicit smoothing priors inherent in such unsupervised deep generative models [41], which can substantially suppress noise components during optimization. However, this excessive smoothing tendency often leads to weakened details and loss of textures. On the contrary, although GSMC yields comparatively lower SNR values, they are consistently higher than those of the

TABLE I
QUANTITATIVE COMPARISON OF MTF COMPENSATION PERFORMANCE ACROSS FIVE SCENES

Metric		Degraded image	GSMC (Ours)	Wiener	SelfDeblur	DIPDKP	W-DIP	IMTF
Scene1	SNR (↑)	25.01	25.81	25.71	30.98	**35.39**	28.72	28.74
	RMS (↑)	0.0997	**0.1171**	0.0997	0.1121	0.1116	0.1154	0.0997
	ESL (↑)	217.48	**369.02**	249.62	312.99	223.23	293.29	296.47
Scene2	SNR	24.64	26.77	25.21	37.89	26.83	**43.23**	28.49
	RMS	0.101	**0.1203**	0.101	0.1133	0.1024	0.1176	0.101
	ESL	158.28	**247.99**	172.98	193.47	138.48	182.77	206.39
Scene3	SNR	23.92	25.66	27.56	37.08	26.43	**44.66**	29.77
	RMS	0.1382	0.1473	0.1383	0.1443	0.1418	**0.1476**	0.1383
	ESL	110.93	**196.01**	121.72	145.23	104.66	144.29	150.28
Scene4	SNR	26.13	41.7	39.26	64.27	42.17	**67.05**	31.66
	RMS	0.2665	0.2789	0.2665	0.2784	0.2721	**0.2853**	0.2665
	ESL	113.35	**220.87**	193.17	178.47	92.15	188.31	161.53
Scene5	SNR	16.67	18.42	17.77	24.4	28.82	**25.49**	19.78
	RMS	0.0541	0.0579	0.0541	0.0558	0.0578	**0.0585**	0.0541
	ESL	57.16	**86.72**	64.75	63.71	59.91	65.31	79.74
Time consumption(s)		-	239	**0.6**	221	305	492	12

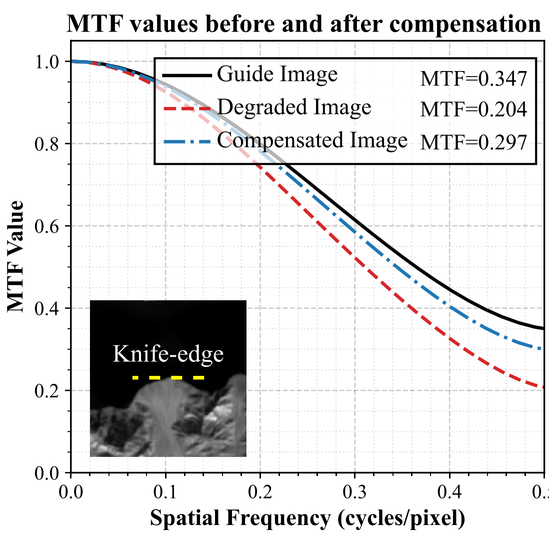

Fig. 7. MTF curves of the guide, degraded, and compensated images.

original degraded images. While maintaining a reasonable SNR level, GSMC more effectively preserves and enhances edge structures and local contrast. For geological remote sensing applications, the fidelity of textures, linear features, and subtle structural information is generally more critical than noise suppression alone.

B. Improvement of MTF and Other Indicators

To quantitatively evaluate the restoration performance of GSMC in the frequency domain, a region with pronounced radiometric discontinuities at the water–land boundary of a lakeshore was selected. The MTF curves were computed and plotted using the knife-edge method. The experimental results are shown in Fig. 7. It can be observed that the MTF curve of the original degraded image (red dashed line) exhibits sharp attenuation even in the low-frequency range, indicating severe loss of edge information. On the contrary, the MTF curve of the image compensated by the GSMC method (blue dash-dotted line) shows substantial improvement. The MTF at the Nyquist frequency increases from 0.204 to 0.297 after restoration. This result confirms that GSMC functions as an effective MTFC method in the strict sense, achieving physically meaningful restoration of the system MTF rather than merely enhancing image sharpness at the perceptual level.

To quantitatively evaluate the noise amplification introduced by MTF compensation in the frequency domain, the noise power spectrum (NPS) was computed for several relatively homogeneous regions within the study area using the outputs of different methods, as shown in Fig. 8. Overall, in most scenes (Scenes 1–3 and 5), the NPS curves exhibit the typical monotonically decreasing trend with increasing spatial frequency. The differences among methods are mainly reflected in their decay behavior in the medium- and high-frequency ranges. Some methods exhibit elevated or slowly decaying high-frequency energy, indicating that additional high-frequency noise is introduced during detail enhancement. On the contrary, the proposed GSMC method preserves a spectral profile similar to that of the degraded imagery, with compensation concentrated primarily in the mid-frequency range and without abnormal amplification of high-frequency components. It is noteworthy that the NPS curve of the water region in Scene 4 exhibits a markedly different pattern from those of the other scenes, characterized by pronounced flattening in the medium- and high-frequency ranges rather than continuous decay. Inspection of the corresponding image reveals weak stripe structures caused by sensor response nonuniformity. This type of noise is strongly directional, and its spectral energy is typically concentrated within specific frequency bands, which leads to a nontypical flattened profile after radial averaging of the NPS. Therefore, the anomalous behavior observed in Scene 4 is not induced by the deblurring algorithms themselves, but rather reflects the dominance of structured noise in the original data. Under such conditions, different methods respond differently to this type of noise.

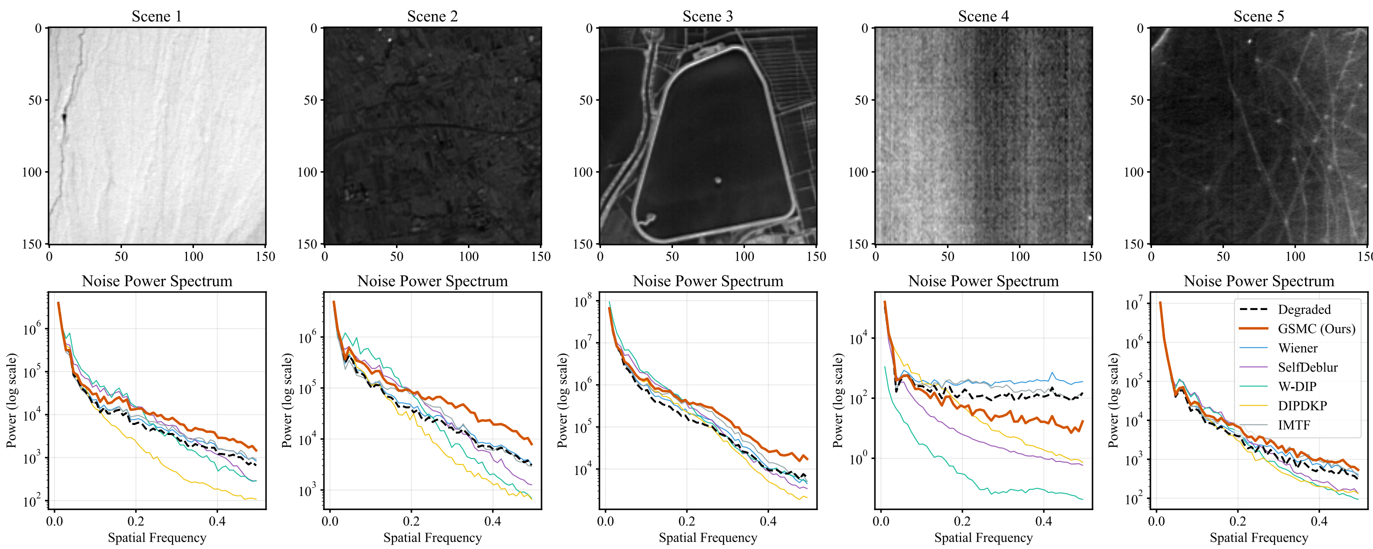

Fig. 8. NPS analysis across five scenes for different MTF compensation methods.

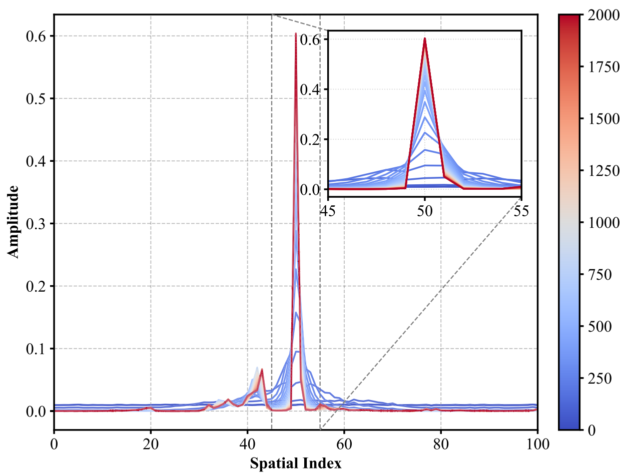

Fig. 9. Evolution of the estimated PSF during Stage 1 optimization.

Some methods, such as W-DIP, exhibit stronger suppression of structured noise, whereas others tend to preserve or partially amplify the energy within these frequency bands. From an overall perspective, the proposed method does not introduce additional abnormal spectral structures across all scenes. The NPS curves remain continuous and smoothly decaying over the entire frequency range, indicating that the MTF compensation process does not produce unstable noise gain.

C. Stability of the Training Process

Fig. 9 illustrates the evolution of the PSF estimation during Stage 1 as the number of iterations increases. The estimated PSF progressively stabilizes, with the peak intensity exceeding 0.5 and the energy becoming increasingly concentrated toward the central pixel. As the iteration count increases from 0 to 2000 (indicated by the color transition from blue to red), the sidelobe noise at the PSF boundaries is gradually suppressed and approaches zero, thereby validating the effectiveness of the proposed PSF estimation network. The local magnified views further reveal that, in the later stages of optimization, the PSF converges to a sharp waveform approximating a Delta function, with both its amplitude and shape remaining highly stable. This behavior demonstrates the favorable convergence properties and robustness of the proposed PSF estimation approach. After PSF convergence, low-amplitude nonzero responses can still be observed in the vicinity of the main peak. This phenomenon primarily arises from residual weak ghosting effects present in the processed imagery, to which the PSF estimation remains responsive. For instance, in the coastal edge region of the port area in Scene 4 shown in Fig. 6, slight ghosting structures are clearly visible, and their corresponding blur characteristics manifest in the PSF domain as low-amplitude responses surrounding the main peak.

Fig. 10 illustrates the dynamic evolution of the latent sharp image reconstruction network during Stage 2 under the condition of a known PSF, as the number of iterations increases. The optimization convergence curve in the upper right corner shows that the loss function decreases rapidly during the initial phase (the first 20 iterations), indicating that the network is able to quickly reconstruct the dominant structural information of the image. Subsequently, the loss enters a stable plateau after approximately 50 iterations and exhibits only low-amplitude fluctuations in the later iterations, demonstrating the high convergence efficiency and favorable numerical stability of the adopted optimization strategy. This numerical convergence behavior is visually confirmed by the local magnified results shown below. In the early iterations, the compensated images still contain noticeable artifacts; as the iteration count increases to 50 and 100, the outlines of ground objects become progressively clearer. In particular, for the selected high-contrast regions such as the water–land boundary along the lakeshore, the diffusive blur present in the original imagery is continuously suppressed. By the final iteration (Iter 200), the boundary between water and land appears sharp and well-defined, without the ringing artifacts commonly observed in traditional deconvolution methods, and both texture details and edge contrast are effectively compensated.

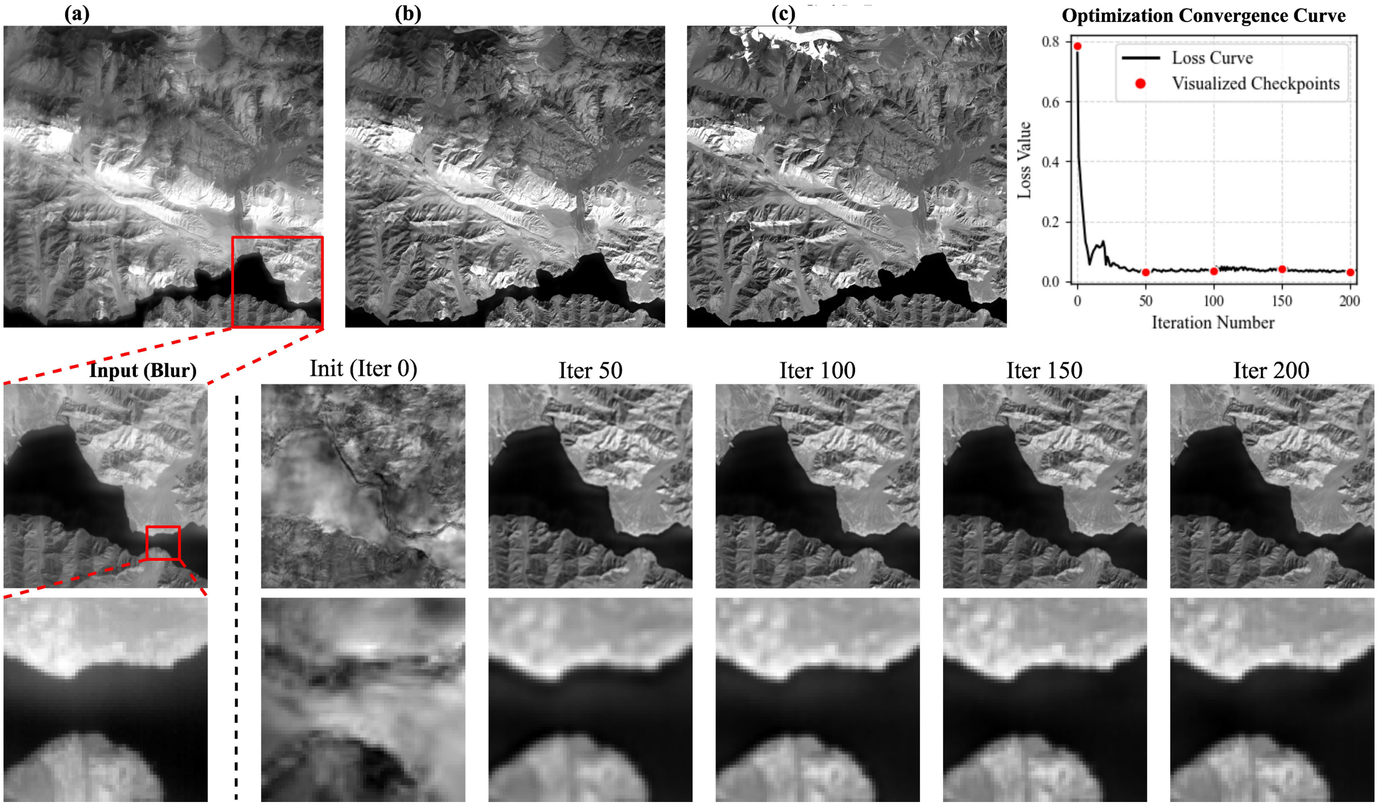

Fig. 10. Progressive restoration results at different iteration steps in Stage 2. (a) Degraded image. (b) Compensated image. (c) Guide image.

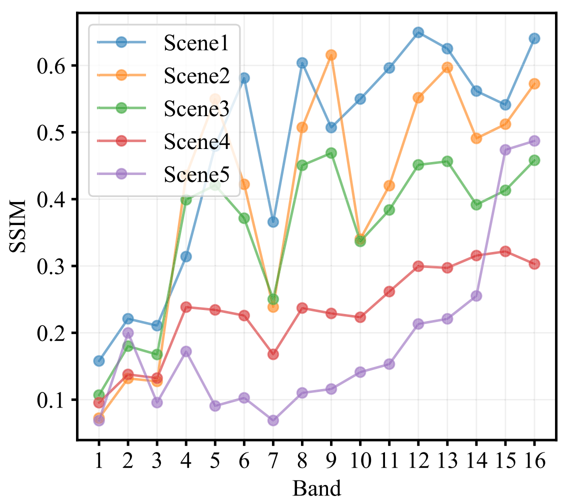

Fig. 11. Comparison of SSIM values between different VNIR spectral bands and SWIR B6 imagery across five scenes.

D. Different VNIR Guided Images for Compensation

Owing to the substantial differences in radiometric energy distribution and imaging mechanisms between VNIR and SWIR imagery, the structural consistency between the selected guidance band and the target SWIR imagery directly affects PSF estimation and compensation performance. As shown in Fig. 11, the SSIM was computed between 16 VNIR spectral bands and the SWIR B6 imagery across five representative scenes. The results indicate that structural similarity between VNIR and SWIR generally increases with wavelength across the VNIR range, suggesting that longer wavelength near-infrared bands exhibit stronger structural consistency with SWIR. Furthermore, Fig. 12 presents the PSF estimation results and corresponding compensated images obtained using different guidance bands, with Scene 4 taken as an example. When B11–B16 (near-infrared region) are used as guidance, the estimated PSFs are more concentrated and stable, and the compensated images show higher structural consistency with the SWIR imagery, particularly at object boundaries such as the water–land interface. Among these bands, B15 and B16 produce the most favorable results. On the contrary, when visible-wavelength bands (B1–B10) are used as guidance, the larger structural discrepancies cause noticeable deviations and instability in PSF estimation, which in turn introduce evident ripple-like artifacts into the compensation results. Overall, the structural similarity between the guidance image and the target imagery is a key factor governing compensation quality. Selecting near-infrared bands that are spectrally adjacent to SWIR can significantly improve the stability of PSF estimation and the reliability of the compensation results.

E. Radiometric Fidelity of Restored Images

Fig. 13 verifies, from a radiometric perspective, the ability of the proposed algorithm to preserve the original digital number (DN) values while performing MTF compensation. A complex scene containing diverse land-cover types, including water bodies, cropland, and impervious surfaces, is selected for evaluation. To analyze radiometric variation trends in the spatial domain, six profile lines are uniformly sampled along

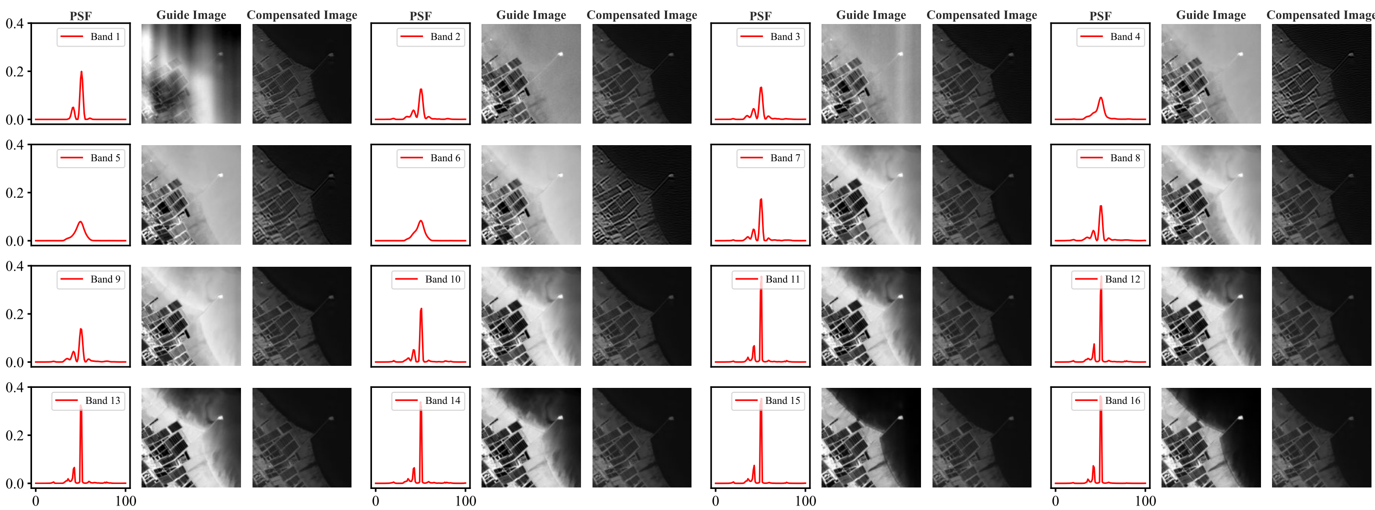

Fig. 12. Comparison of PSF estimation and compensation results using different VNIR spectral bands as guidance (Scene 4).

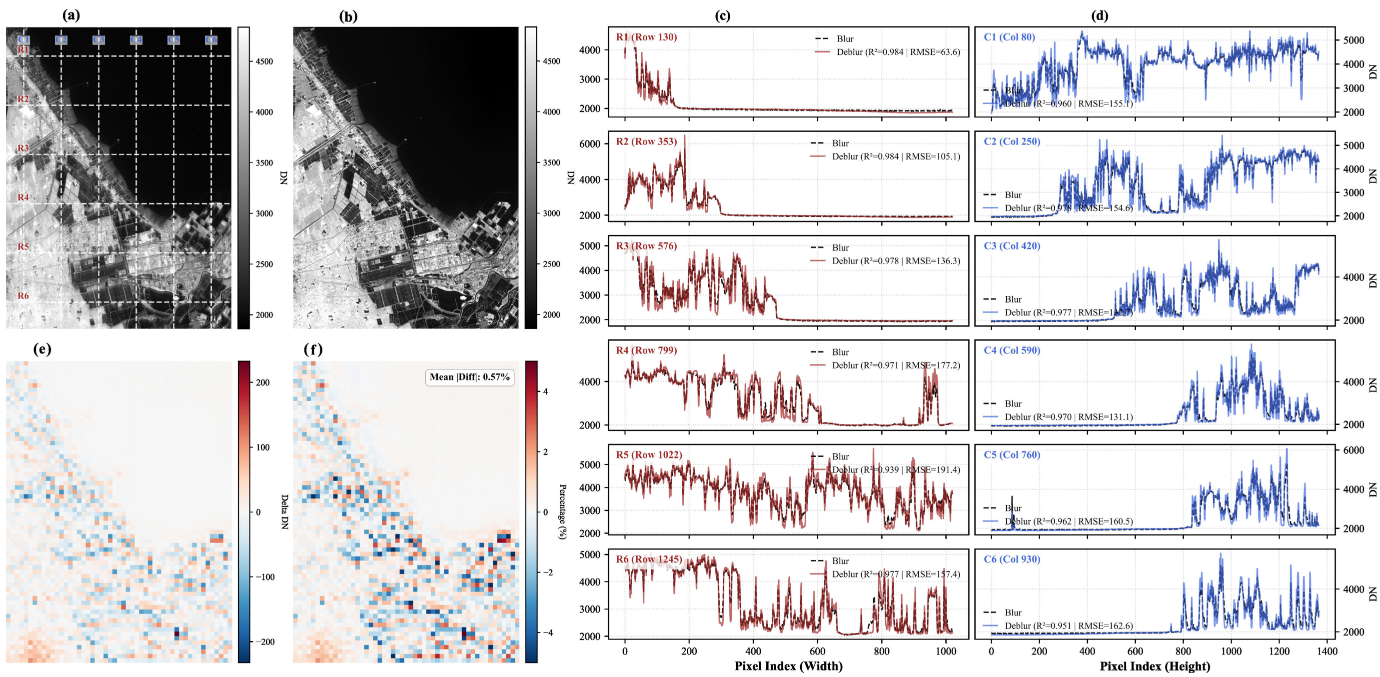

Fig. 13. Radiometric fidelity assessment of the proposed GSMC method. (a) Degraded image, (b) compensated image, (c) DN variation in the along-track direction, (d) DN variation in the cross-track direction, (e) absolute DN difference, and (f) relative difference rate.

both the along-track and cross-track directions (R1–R6 and C1–C6), as shown in Fig. 13(c) and (d). As indicated by the waveform comparisons, the DN curves of the compensated image (solid lines) closely coincide with those of the original degraded image (dashed lines) in terms of overall trends, faithfully following the radiometric variations of ground objects. Quantitative statistics show that the coefficient of determination (R^2) for all profiles exceeds 0.95, while the root-mean-square error (RMSE) is constrained within 200, demonstrating that the proposed algorithm does not disrupt the inherent low-frequency radiometric structure of the image. Furthermore, Fig. 13(e) and (f) presents the spatial distribution of DN differences before and after restoration using a 20×20 pixel sliding window. The absolute difference heatmap in Fig. 13(e) indicates that, except for a few strong edge regions, DN fluctuations are minimal over most areas. Fig. 13(f) further quantifies this fidelity, showing that the mean absolute value of the relative error rate over the entire image is only 0.57%. Two representative downstream aspects are further examined to evaluate the practical benefits of the compensated imagery.

1) Quantitative Remote Sensing Inversion: To verify whether the cross-spectral structural prior introduces spectral distortion into SWIR-specific features, particularly in geological scenarios where SWIR-dependent physical characteristics are pronounced, a targeted experiment was conducted over the Haoyaerhudong gold deposit in the Xinhuore area of Inner Mongolia, as shown in Fig. 14(a). As illustrated in Fig. 14(a), the epidote–chlorite–amphibole index (ECAI) derived from Geology-1 data indicates that mineralization anomalies are already discernible in the degraded imagery, whereas the

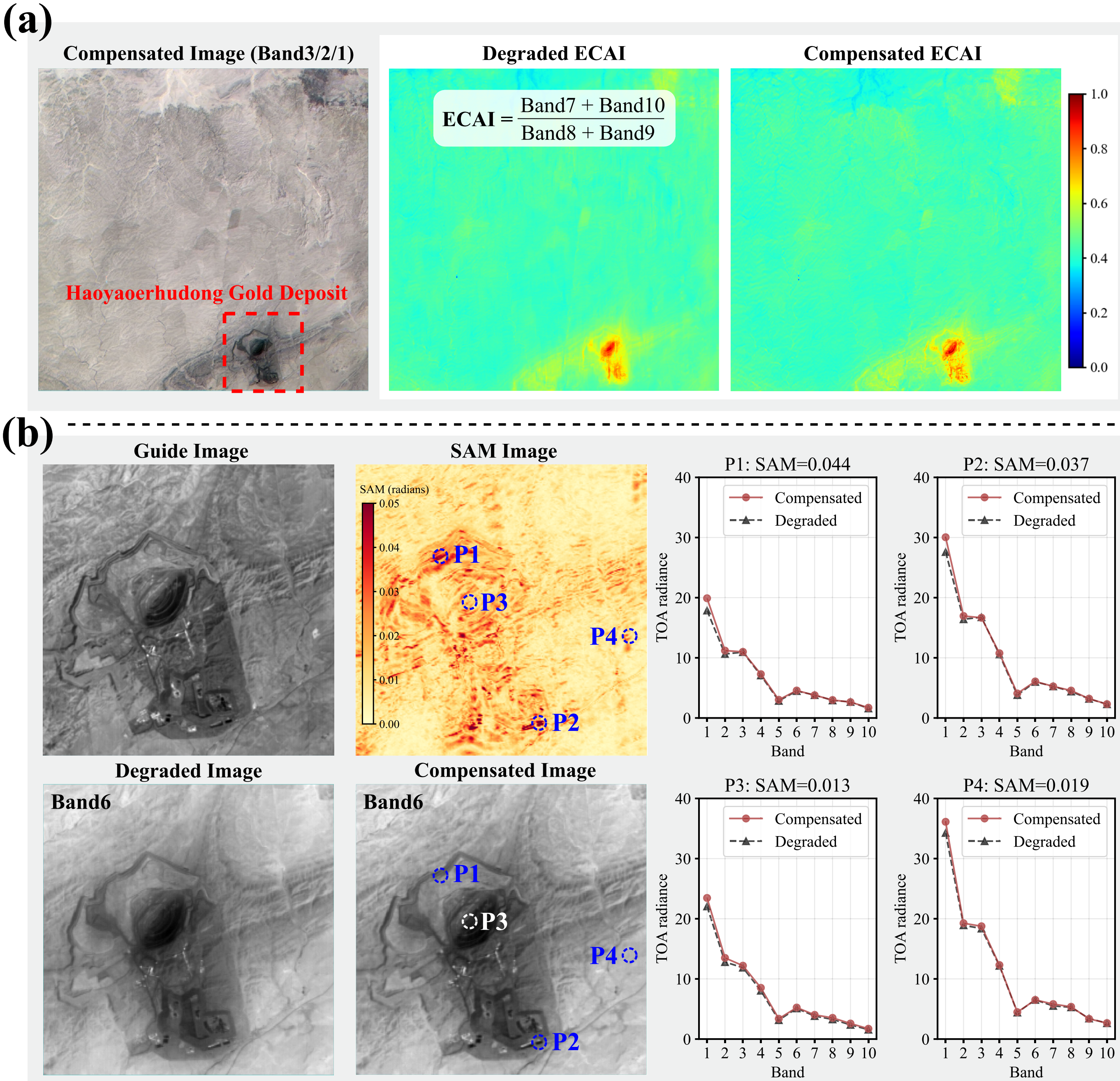

Fig. 14. Validation of cross-spectral structural constraint using ECAI response and spectral consistency in a gold deposit area. (a) Comparison of ECAI maps derived from degraded and compensated SWIR imagery, demonstrating the preservation of mineralization anomaly patterns after compensation. (b) SAM-based spectral consistency analysis and spectral profile comparison at representative pixels (P1–P4), showing that the compensated imagery maintains the characteristic SWIR absorption features.

compensated imagery exhibits clearer spatial structures and higher contrast. No abnormal changes in anomaly morphology or spatial distribution are observed, suggesting that the compensation process preserves the mineral spectral information governed by SWIR-specific reflectance characteristics. Furthermore, spectral consistency analysis was performed over locally enlarged regions of the mining area, as shown in Fig. 14(b). The spectral angle mapper (SAM) was employed to quantitatively assess spectral differences before and after compensation. SAM values at representative pixel locations are all below 0.05, indicating a high degree of spectral-shape consistency. For pixels with relatively larger deviations, profile analysis confirms that the compensated results do not exhibit filling-in or attenuation of the characteristic SWIR absorption features. Taken together, the index response and spectral angle analyses demonstrate that the adopted cross-spectral structural constraint effectively enhances spatial detail without introducing significant suppression or distortion of the characteristic geological spectral information in the SWIR domain.

2) Land-Cover Classification: To verify the performance gains of GSMC in practical remote sensing applications, we apply both the original and compensated imagery to a downstream land-cover classification task. A feature space consisting of 14 spectral dimensions is constructed as input, including VNIR bands (B2-Blue, B5-Green, B7-Red, B14-NIR) and ten SWIR bands. A support vector machine (SVM) classifier is employed to categorize the study area into four land-cover classes: water, impervious surface, bare soil, and cropland. Fig. 15(a) and (b) presents the classification confidence maps generated from the original and compensated imagery, respectively. The confidence maps provide a fine-grained representation of the classifier's confidence at the pixel level, thereby reflecting the influence of data quality on feature separability. Quantitative evaluation shows that classification accuracy based on the compensated imagery is substantially improved, with the overall accuracy increasing from 83.43% to 89.19%. This improvement is primarily attributed to the MTF compensation process, which effectively reduces spectral overlap between adjacent pixels. From a visual perspective, as illustrated in the magnified regions, the motion-induced smearing in the original imagery leads to a large number of mixed pixels along land-cover boundaries, resulting in markedly reduced classification confidence in these areas, manifested as cooler colors or low-value regions. On the contrary, the results obtained from the compensated imagery exhibit the following advantages.

1) *Enhanced Boundary Certainty:* The confidence levels along land-cover edges, such as the boundaries between cropland and roads, are significantly increased, indicating that the proposed method effectively sharpens object contours and reduces boundary ambiguity caused by blur.
2) *Improved Detail Discriminability:* The classification results show higher heterogeneity and finer spatial granularity. This suggests that small-scale land-cover features were previously smoothed out by blur, such as small impervious surfaces embedded within large bare-land areas, are successfully identified and separated, thereby substantially enhancing the level of detail in remote sensing mapping.

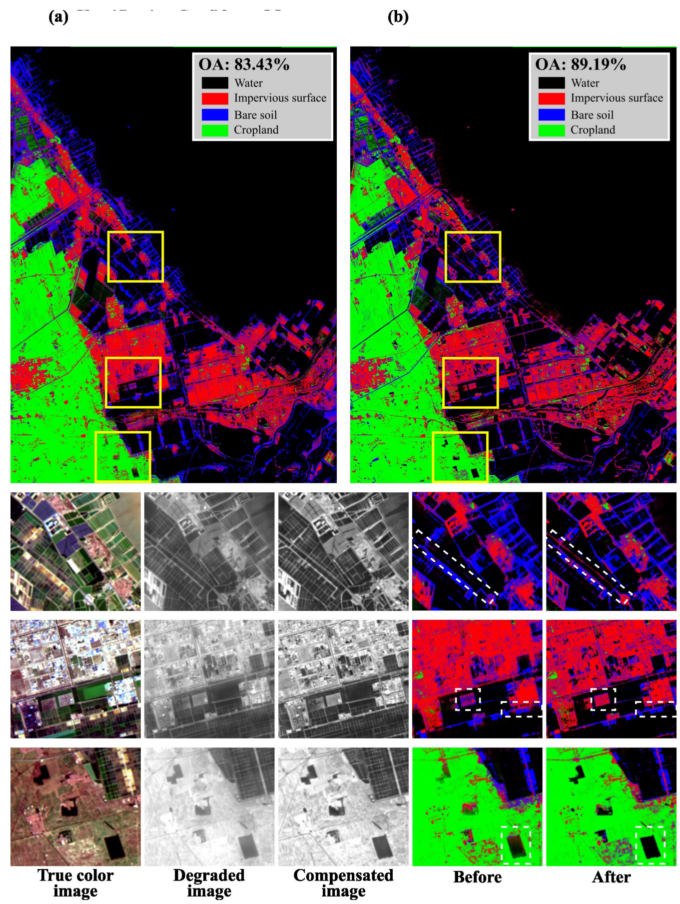

Fig. 15. Land-cover classification confidence maps before and after MTF compensation. (a) Classification confidence map based on degraded bands (before). (b) Classification confidence map based on compensated bands (after).

F. Engineering Applications of GSMC

We estimated the equivalent PSF of 80 SWIR images acquired in different regions within the same spectral band and performed statistical analysis on all estimated PSFs. Further statistical results indicate that, for most degraded images, the estimated PSFs exhibit a high degree of consistency with the mean PSF in terms of the main-peak amplitude and energy distribution characteristics. In particular, the energy proportion within the main-peak neighborhood of most samples falls within the statistical fluctuation range of the mean PSF, with only a small number of images showing relatively noticeable deviations in sidelobe amplitudes. This suggests that although the PSF exhibits certain temporal variability at the scene level, such variations are primarily confined to secondary energy distribution regions, while the dominant structures that determine the degradation characteristics remain relatively stable within the same spectral band.

We further investigate the feasibility of replacing per-scene PSF estimation with a band-level mean PSF in practical engineering applications, and apply this strategy to the non-blind MTF compensation process in Stage 2. Experimental results show that, while maintaining overall consistency in restoration performance, this approach can significantly reduce computational cost. In particular, the average processing time per scene decreases from 239 s, which includes the PSF estimation stage, to approximately 34 s, yielding a sevenfold improvement in processing efficiency.

To quantify the error bounds introduced by this engineering simplification, we compare the MTF recovery metrics obtained using the band-level mean PSF (referred to as the robust kernel) against those obtained with per-scene PSF estimation. The first three subplots in Fig. 16 show that, across five representative scenes, the robust kernel achieves restoration performance comparable to that of the scene-specific kernels in terms of SNR, rms, and ESL. For example, the maximum SNR deviation between the robust kernel and the scene-specific kernels is less than 2 dB across all scenes, while the deviations in rms and ESL remain within 5% and 8%, respectively. The fourth subplot further shows that the robust kernel preserves the main-peak structure of the scene-specific kernels, with sidelobe amplitude deviations below 5% in the majority of scenes. These results demonstrate that, within the same spectral band, a statistically representative PSF approximation can effectively balance computational efficiency and restoration accuracy, thereby providing a practical basis for the large-scale operational deployment of the GSMC method in remote sensing data processing.

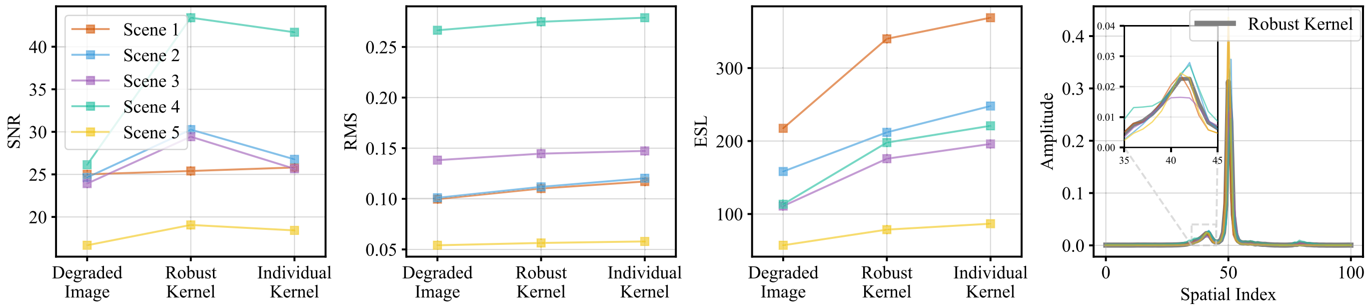

Fig. 16. Comparison of MTF recovery performance between robust kernel and per-scene individual kernels across five scenes.

IV. Conclusion

This study addresses the MTF degradation of SWIR imagery encountered during the on-orbit operation of the Geology-1 hyperspectral satellite by proposing a VNIR structure-guided and zero-shot blind MTF compensation framework. By exploiting structural priors from the coregistered VNIR bands of the same sensor, the proposed approach enables robust estimation of time-varying and spatially nonuniform PSFs in the absence of paired degraded-sharp reference data. Experimental results demonstrate that this cross-spectral structure transfer mechanism effectively bridges the representational gap between spatial and spectral features, allowing accurate reconstruction of lost high-frequency geological textures. Quantitative evaluations indicate that the GSMC framework increases the MTF at the Nyquist frequency from 0.204 to 0.297. High radiometric fidelity is maintained throughout, with a mean relative error of only 0.57%. In downstream land-cover classification tasks, the compensated imagery improves overall accuracy from 83.43% to 89.19%. Furthermore, spectral consistency analysis confirms that the cross-spectral guidance strategy does not distort SWIR-specific spectral features, with SAM values remaining below 0.05. In addition, the proposed band-level mean PSF strategy achieves a sevenfold computational speedup while maintaining restoration accuracy.

Although this study achieves notable progress in zero-shot MTF compensation, there remains room for further improvement. Future work will first seek to integrate more comprehensive physical imaging models by constructing a complete physical blur chain that accounts for atmospheric coherence distortions, optical system diffraction, and detector sampling effects, thereby further enhancing the theoretical upper bound of PSF estimation under complex on-orbit conditions. In addition, future research will explore deeper coupling between deep structural priors and analytical physical models, using physical laws to constrain the solution space of neural networks, to improve generalization performance in extremely low SNR scenarios.

References

[1] L. Zhu, N. Li, and P. R. N. Childs, "Light-weighting in aerospace component and system design," *Propuls. Power Res.*, vol. 7, no. 2, pp. 103–119, Jun. 2018.

[2] K.-P. Bouzoukis, G. Moraitis, V. Kostopoulos, and V. Lappas, "An overview of CubeSat missions and applications," *Aerospace*, vol. 12, no. 6, p. 550, Jun. 2025.

[3] Y. Song, D. Gnyawali, and L. Qian, "From early curiosity to space wide web: The emergence of the small satellite innovation ecosystem," *Res. Policy*, vol. 53, no. 2, Mar. 2024, Art. no. 104932.

[4] B. Yost and S. Weston, "State-of-the-art small spacecraft technology," NASA Ames Res. Center, Mountain View, CA, USA, Tech. Rep. NASA/TP-20240001462, 2024.

[5] European Space Agency. (2025). *Planetscope Mission Description*. [Online]. Available: https://earth.esa.int/eogateway/missions/planetscope/description

[6] Changguang satellite technology co. (2025). *Changguang Satellite Technology Co., Ltd. Has Developed a Mass Production Technology for Ultra-Lightweight Sub-Meter-Level Optical Satellite Constellations, Contributing To China's Aerospace Industry*. [Online]. Available: https://kjt.jl.gov.cn/xwzx/kjcg/202508/t20250807_9299534.html

[7] L. Wang et al., "Geology-1: The first hyperspectral remote sensing microsatellite in China for geological resource and environmental survey and monitoring [Space-Agencies]," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 4, pp. 455–492, Dec. 2025.

[8] V. Preda, "Robust microvibration control and worst-case analysis for high pointing stability space missions," Ph.D. dissertation, Université de Bordeaux, Talence, France, 2017.

[9] D. Llaveria, A. Camps, and H. Park, "Correcting the ADCS jitter induced blurring in small satellite imagery," *IEEE J. Miniaturization for Air Space Syst.*, vol. 1, no. 2, pp. 130–137, Sep. 2020.

[10] Z. Feng, J. Chang, Y. Hu, Z. Liu, and Y. Cui, "Micro-vibration modeling and suppression technology and prospects for high resolution remote sensing satellite," in *Proc. Int. Symp. Space Opt. Instrum. Appl.*, 2023, pp. 535–546.

[11] I. Kazemi, M. Abolghasemi Najafabadi, and M. Shafiee, "Enhancing small satellite optical imaging performance under resolution limitations in diffraction, motion, and noise," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–14, 2025.

[12] S. Liu et al., "Attitude jitter detection based on remotely sensed images and dense ground controls: A case study for Chinese ZY-3 satellite," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 12, pp. 5760–5766, Dec. 2016.

[13] J. Li, Z. Liu, and S. Liu, "Suppressing the image smear of the vibration modulation transfer function for remote-sensing optical cameras," *Appl. Opt.*, vol. 56, no. 6, pp. 1616–1624, 2017.

[14] X. Zhu, S. Cohen, S. Schiller, and P. Milanfar, "Estimating spatially varying defocus blur from a single image," *IEEE Trans. Image Process.*, vol. 22, no. 12, pp. 4879–4891, Dec. 2013.

[15] M. Bauer, V. Volchkov, M. Hirsch, and B. Schcölkopf, "Automatic estimation of modulation transfer functions," in *Proc. IEEE Int. Conf. Comput. Photography (ICCP)*, May 2018, pp. 1–12.

[16] S. De Lellis, "A methodology to account for dynamic variability in micro-vibration analysis of satellites," Ph.D. dissertation, University of Surrey, Guildford, U.K., 2020.

[17] J. Han, S. Zhang, and Z. Ye, "A nonblind deconvolution method by bias correction for inaccurate blur kernel estimation in image deblurring," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604814.

[18] M. S. Hosseini and K. N. Plataniotis, "Convolutional deblurring for natural imaging," *IEEE Trans. Image Process.*, vol. 29, pp. 250–264, 2020.

[19] A. Karaali and C. R. Jung, "Edge-based defocus blur estimation with adaptive scale selection," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1126–1137, Mar. 2018.

[20] L. Li, J. Cao, S. Wei, Y. Jiang, and X. Shen, "Improved on-orbit MTF measurement method based on point source arrays," *Remote Sens.*, vol. 15, no. 16, p. 4028, Aug. 2023.

[21] J. Rong, H. Huang, and J. Li, "IMU-assisted accurate blur kernel re-estimation in non-uniform camera shake deblurring," *IEEE Trans. Image Process.*, vol. 33, pp. 3823–3838, 2024.
[22] O. Beltramo-Martin et al., "Review of PSF reconstruction methods and application to post-processing," *Proc. SPIE*, vol. 11448, pp. 22–36, Dec. 2020.
[23] Z. Song, Z. Zhang, F. Fang, Z. Fan, and J. Lu, "Deep semantic-aware remote sensing image deblurring," *Signal Process.*, vol. 211, Oct. 2023, Art. no. 109108.
[24] A. Valenzuela, K. Reinke, and S. Jones, "A new procedure to find the optimum deconvolution kernel to deblur satellite images," *Int. J. Remote Sens.*, vol. 45, no. 24, pp. 9513–9537, Dec. 2024.
[25] O. Ozan, G. Kaya, G. Tarhan, M. Doğan, and N. G. Kasapoğlu, "Performance comparison of deblurring techniques for optical satellite imagery using MTF," in *Proc. 10th Int. Conf. Recent Adv. Air Space Technol. (RAST)*, Jun. 2023, pp. 1–5.
[26] J. Dong, S. Roth, and B. Schiele, "Deep Wiener deconvolution: Wiener meets deep learning for image deblurring," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 1048–1059.
[27] F. Ji, J. Wang, S. Cui, J. Li, X. Tang, and F. Xu, "Physics-guided optical simulation and PSF analysis for remote sensing images deblurring," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5633015.
[28] D. Ren, K. Zhang, Q. Wang, Q. Hu, and W. Zuo, "Neural blind deconvolution using deep priors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3338–3347.
[29] G. Bredell, E. Erdil, B. Weber, and E. Konukoglu, "Wiener guided DIP for unsupervised blind image deconvolution," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 3046–3055.
[30] Z. Yang et al., "A dynamic kernel prior model for unsupervised blind image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 26046–26056.
[31] L. Zheng, Z. Yang, and J. Xia, "Unsupervised hyperspectral image super-resolution via learning kernel distributions in alternating framework," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Aug. 2025, pp. 8910–8914.
[32] J. Xia et al., "Blind super-resolution via meta-learning and Markov chain Monte Carlo simulation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 8139–8156, Dec. 2024.
[33] D. Scheffler, A. Hollstein, H. Diedrich, K. Segl, and P. Hostert, "AROSICS: An automated and robust open-source image co-registration software for multi-sensor satellite data," *Remote Sens.*, vol. 9, no. 7, p. 676, Jul. 2017.
[34] D. Kundur and D. Hatzinakos, "Blind image deconvolution," *IEEE Signal Process. Mag.*, vol. 13, no. 3, pp. 43–64, May 1996.
[35] F. Sroubek and J. Flusser, "Multichannel blind iterative image restoration," *IEEE Trans. Image Process.*, vol. 12, no. 9, pp. 1094–1106, Sep. 2003.
[36] D. Krishnan, T. Tay, and R. Fergus, "Blind deconvolution using a normalized sparsity measure," in *Proc. CVPR*, Jun. 2011, pp. 233–240.
[37] K. Rani, "Mineral exploration using remote sensing," in *Remote Sensing for Geophysicists*. Boca Raton, FL, USA: CRC Press, 2025, pp. 79–94.
[38] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*, 2015, pp. 234–241.
[39] K. Kondo, Y. Ichioka, and T. Suzuki, "Image restoration by Wiener filtering in the presence of signal-dependent noise," *Appl. Opt.*, vol. 16, no. 9, pp. 2554–2558, 1977.
[40] L. Qi et al., "Fast thermal infrared image restoration method based on on-orbit invariant modulation transfer function," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5001015.
[41] V. Lempitsky, A. Vedaldi, and D. Ulyanov, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9446–9454.

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